

OVERKALIX-SVARTBYN, Sweden

WMO: 021810

Lat: 66.262N

Lon: 22.843E

Elev: 62

StdP: 100.58

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.6	-27.5	-34.0	0.2	-30.6	-30.6	0.2	-27.4	7.3	-2.1	6.5	-2.3	0.2	330	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	25.5	17.2	23.6	16.3	21.7	15.5	18.7	23.2	17.5	21.7	16.4	20.2	2.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	12.3	20.3	15.8	11.3	19.0	14.7	10.6	18.1	53.2	23.0	49.4	21.7	46.3	20.3	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.2	5.3	4.5	-34.0	28.1	3.9	2.2	-36.8	29.7	-39.0	31.0	-41.2	32.3	-44.0	33.9		
			-34.2	20.2	3.8	1.2	-36.9	21.1	-39.1	21.8	-41.2	22.4	-44.0	23.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.4	-11.8	-11.0	-6.1	0.4	6.9	12.5	15.9	13.5	8.3	1.4	-5.0	-9.1
	DBStd	10.97	8.09	7.91	5.88	3.87	3.95	3.38	3.02	3.07	3.60	4.56	6.31	7.61
	HDD10.0	3567	674	589	498	289	113	13	1	7	74	266	449	594
	HDD18.3	6199	933	822	756	539	356	177	88	153	301	524	699	852
	CDD10.0	425	0	0	0	0	17	88	183	114	23	0	0	0
	CDD18.3	15	0	0	0	0	0	2	12	2	0	0	0	0
	CDH23.3	191	0	0	0	0	10	28	133	19	0	0	0	0
	CDH26.7	26	0	0	0	0	1	23	1	0	0	0	0	0
Wind	WSAvg	1.8	1.5	1.8	2.1	2.0	2.2	2.1	1.8	1.7	1.8	1.8	1.6	1.6
Precipitation	PrecAvg	571	40	34	30	30	39	55	73	63	56	47	53	46
	PrecMax	784	79	73	59	77	122	87	117	140	130	92	101	111
	PrecMin	422	6	5	10	7	5	8	26	15	15	13	18	7
	PrecStd	98	17	19	16	19	23	23	24	28	29	22	23	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	5.8	7.9	14.4	24.4	26.6	29.2	25.4	18.9	11.6	7.1	4.4
		MCWB	2.8	3.2	3.9	8.1	15.2	16.9	19.0	17.8	14.0	9.0	5.7	2.7
	2%	DB	1.6	2.5	5.5	11.0	20.3	23.9	26.6	23.4	17.0	10.2	5.2	2.3
		MCWB	0.5	0.9	2.1	5.8	12.5	15.2	17.9	16.9	12.8	8.8	4.5	1.7
	5%	DB	0.3	0.7	3.6	8.6	17.0	21.8	24.6	21.5	15.5	9.0	3.6	1.3
		MCWB	-0.1	0.0	1.0	4.4	10.4	14.2	17.3	16.0	12.3	7.9	3.1	0.8
	10%	DB	-0.8	-0.4	1.6	6.6	14.5	19.6	22.6	19.5	14.1	7.5	2.3	0.5
		MCWB	-1.3	-0.9	-0.1	3.1	8.9	13.2	16.4	15.2	11.6	6.5	1.9	0.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	3.4	4.2	8.3	15.6	18.3	20.7	19.5	15.0	10.2	6.2	3.1
		MCDB	4.6	5.5	7.7	13.7	22.8	24.1	26.2	23.4	17.1	10.8	6.6	3.9
	2%	WB	0.8	1.2	2.5	6.2	13.1	16.5	19.3	18.0	13.9	9.1	4.6	1.9
		MCDB	1.3	2.1	5.0	10.2	18.6	21.5	24.2	21.4	15.9	9.9	5.2	2.3
	5%	WB	0.0	0.1	1.2	4.7	11.1	15.1	18.1	16.8	13.0	7.9	3.2	1.0
		MCDB	0.3	0.7	2.8	8.0	16.1	20.1	22.7	19.9	14.7	8.9	3.6	1.3
	10%	WB	-1.3	-0.9	0.3	3.4	9.5	14.0	17.1	15.7	12.1	6.6	1.9	0.1
		MCDB	-0.8	-0.4	1.4	6.2	13.7	18.3	21.3	18.8	13.6	7.5	2.3	0.4
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.3	9.7	9.1	9.9	9.5	9.8	9.4	8.2	5.8	5.7	7.3
		MCDBR	6.8	7.0	9.9	12.1	15.0	13.6	13.7	12.5	9.9	6.8	4.6	5.3
		MCWBR	6.5	6.1	7.4	7.5	7.9	6.2	6.2	6.7	6.3	5.4	4.2	5.0
	5% WB	MCDBR	6.3	6.6	8.6	11.3	13.9	11.5	11.5	10.6	8.1	6.0	4.4	4.9
		MCWBR	6.0	5.9	6.6	7.3	7.7	6.0	6.1	5.5	5.1	4.1	4.1	4.7
Clear-Sky Solar Irradiance	taub	0.198	0.267	0.298	0.332	0.343	0.346	0.359	0.343	0.315	0.291	0.341	0.355	
	taud	1.944	2.125	2.200	2.280	2.452	2.474	2.454	2.488	2.545	2.373	2.649	2.674	
	Ebn at Noon	224	581	757	827	861	865	838	817	763	572	234	65	
	Edh at Noon	17	53	83	99	93	94	93	81	61	43	17	8	
All-Sky Solar Radiation	RadAvg	0.08	0.60	2.10	3.86	4.91	5.36	5.01	3.67	1.99	0.72	0.14	0.01	
	RadStd	0.01	0.10	0.20	0.23	0.49	0.42	0.46	0.31	0.27	0.13	0.02	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page